

EXAM 7 – SPRING 2015

5. (2.5 points)

Given the following information:

Age-to-Age Loss Development Factors

Accident Year	12-24 Months	24-36 Months	36-48 Months	48-60 Months	60-72 Months
2009	2.50	1.45	1.06	1.05	1.01
2010	1.80	1.38	1.15	1.02	
2011	2.13	1.20	1.13		
2012	2.18	1.36			
2013	1.95				

Degrees of Freedom	<i>t</i> -statistic for 0.9
1	3.078
2	1.886
3	1.638
4	1.533
5	1.476
6	1.440

Assume that $T = r \cdot [(n - 2) / (1 - r^2)]^{1/2}$ is *t*-distributed with $(n - 2)$ degrees of freedom, where r is the sample correlation coefficient.

Using Mack's correlation test with a 10% *t*-statistic significance standard, test the null hypothesis that the 12-24 month and 24-36 month age-to-age factors are independent.

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QUESTION 5

TOTAL POINT VALUE: 2.5

LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as chain ladder, Cape Cod, chain ladder plus calendar-year effects, and Bornhuetter-Ferguson

SAMPLE ANSWERS

Sample Answer 1

AY	X	Y	$(X - E[X])$	$(Y - E[Y])$	$(X - E[X])^2$	$(Y - E[Y])^2$	$(X - E[X]) \cdot (Y - E[Y])$
2009	1.5000	0.4500	0.3475	0.1025	0.1208	0.0105	0.0356
2010	0.8000	0.3800	(0.3525)	0.0325	0.1243	0.0011	(0.0115)
2011	1.1300	0.2000	(0.0225)	(0.1475)	0.0005	0.0218	0.0033
2012	1.1800	0.3600	0.0275	0.0125	0.0008	0.0002	0.0003
Mean	1.1525	0.3475		Σ	0.2463	0.0335	0.0278

$$r = \frac{\Sigma(X - E[X]) * (Y - E[Y])}{(\Sigma(X - E[X])^2 * \Sigma(Y - E[Y])^2)^{.5}} = \frac{0.0278}{(0.2463 * 0.0335)^{.5}} = 0.3065$$

$$n = 4$$

$$T = r * [(n - 2) / (1 - r^2)]^{.5} = 0.3065 * [(4 - 2) / (1 - .3065^2)]^{.5} = 0.4553$$

t-statistic = 1.8860

Since $0.4553 < 1.8860$, the null hypothesis that the 12-24 month and 24-36 month age-to-age factors are independent is met

Sample Answer 2

AY	X	Y	XY	$(X - E[X])^2$	$(Y - E[Y])^2$
2009	1.5000	0.4500	0.6750	0.1208	0.0105
2010	0.8000	0.3800	0.3040	0.1243	0.0011
2011	1.1300	0.2000	0.2260	0.0005	0.0218
<u>2012</u>	<u>1.1800</u>	<u>0.3600</u>	<u>0.4248</u>	<u>0.0008</u>	<u>0.0002</u>
Mean	1.1525	0.3475	0.4075	Σ 0.2463	0.0335
			n	4	4
			σ^2	0.0616	0.0084
			σ	0.2481	0.0915

$$r = \frac{E[XY] - E[X] * E[Y]}{\sigma_X * \sigma_Y}$$

$$r = \frac{0.4075 - 1.1525 * .3475}{(0.2481 * 0.0915)} \quad r = .3065$$

$$n = 4$$

$$T = r * [(n - 2) / (1 - r^2)]^{.5} = 0.3065 * [(4 - 2) / (1 - .3065^2)]^{.5} = .4553$$

t-statistic=1.8860

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Since $0.4553 < 1.8860$, the null hypothesis that the 12-24 month and 24-36 month age-to-age factors are independent is met.

Sample Answer 3

AY	X	Y	XY	X ²	Y ²
2009	1.5000	0.4500	0.6750	2.2500	0.2025
2010	0.8000	0.3800	0.3040	0.6400	0.1444
2011	1.1300	0.2000	0.2260	1.2769	0.0400
2012	1.1800	0.3600	0.4248	1.3924	0.1296
Mean	1.1525	0.3475	0.4075	1.3898	0.1291
Mean ²	1.3283	0.1208			

$$r = \frac{E[XY] - E[X] * E[Y]}{((E[X^2] - E[X]^2) * (E[Y^2] - E[Y]^2))^{.5}}$$

$$r = \frac{0.4075 - 1.1525 * .3475}{((1.3898 - 1.3283) * (0.1291 - 0.1208))^{.5}}$$

$$r = 0.3065$$

$$n = 4$$

$$T = r * [(n - 2) / (1 - r^2)]^{.5}$$

$$T = 0.3065 * [(4 - 2) / (1 - .3065^2)]^{.5}$$

$$T = 0.4553$$

$$t\text{-stat} = 1.8860$$

Since $0.4553 < 1.8860$, the null hypothesis that the 12-24 month and 24-36 month age-to-age factors are independent is met.

Sample Answer 4 (accepted response using elements of Mack paper)

AY	X	Y	Rank X	Rank Y	Rank (X-Y) ²
2009	2.5000	1.4500	4	4	-
2010	1.8000	1.3800	1	3	4
2011	2.1300	1.2000	2	1	1
2012	2.1800	1.3600	3	2	1

$$S = \Sigma \quad 6$$

$$r = 1 - \frac{S}{n(n^2 - 1)/6}$$

$$r = 1 - \frac{6}{4 * (4^2 - 1)/6}$$

$$r = 0.400$$

$$n = 4$$

$$T = r * [(n - 2) / (1 - r^2)]^{.5}$$

$$T = 0.400 * [(4 - 2) / (1 - .400^2)]^{.5}$$

$$T = 0.6172$$

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t-statistic = 1.8860

Since $0.6172 < 1.8860$, the null hypothesis that the 12-24 month and 24-36 month age-to-age factors are independent is met.

Sample Answer 5 (accepted response using elements of Mack paper)

AY	X	Y	Rank X	Rank Y	Rank (X-Y) ²
2009	2.5000	1.4500	4	4	-
2010	1.8000	1.3800	1	3	4
2011	2.1300	1.2000	2	1	1
2012	2.1800	1.3600	3	2	1
			S =	Σ	6

$$r = 1 - \frac{S}{n(n^2 - 1)/6}$$

$$r = 1 - \frac{6}{4 * (4^2 - 1)/6}$$

$$r = 0.400$$

$$n = 4$$

$$T = r * [(n - 2) / (1 - r^2)]^{.5}$$

$$T = 0.400 * [(4 - 2) / (1 - .400^2)]^{.5}$$

$$T = 0.6172$$

$$\text{Var [T]} = 1 - \frac{(\# \text{ of AY's} - 2) * (\# \text{ of AY's} - 3)}{2}$$

$$\# \text{ of AY's} : 6$$

$$\text{Var [T]} = 0.167$$

$$\text{Std Dev [T]} = 0.408$$

$$t\text{-statistic} = 1.8860$$

$$\text{Range (+/-)} = 0.7700 = 0.408 * 1.886$$

$$\text{Range } (-0.770, 0.770)$$

Since 0.6172 is within the range the null hypothesis that the 12-24 month and 24-36 month age-to-age factors are independent is met.

EXAMINERS' REPORT

Overall, many candidates performed very well on this question. Candidates needed to know key assumptions of the chain ladder models and how to test these assumptions. The core of the question is determining whether the age-to-age factors are independent.

The question referenced Mack's correlation test by mistake; the intended approach was to use Venter's correlation test. Due to this error, we accepted a variety of responses which used some elements of Mack's correlation test. See Sample Answers 4 and 5 for examples of responses receiving full credit even though they were not the intended responses to the question.